JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 90, NO. B6, PAGES 4639–4646, MAY 10, 1985

Paleomagnetism of Miocene East African Rift Sediments and the Calibration of the Geomagnetic Reversal Time Scale

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Paleomagnetic stratigraphy and K-Ar age determinations are reported for the type section of the middle Miocene Ngorora Formation, found in the Kenya rift valley. The magnetostratigraphy is well correlated to the geomagnetic reversal time scale (GRTS) and spans from the lower part of Chron C5 (9) to Chron C5AB-r (14). K-Ar dates were determined for euhedral sanidines, hand-picked from seven tuff horizons within the Ngorora Formation and the underlying Tiim phonolite flow. These dates can therefore be tied directly to the GRTS. The eight dates fall into three discrete groups averaging 12.5 ± 0.22 Ma (mean and standard deviation of results from four tuffs), 11.6 ± 0.06 Ma (mean and standard deviation from three tuffs), and 10.16 ± 0.38 (average of three analyses from one tuff). We interpret the age groups as resulting from three successive eruptive episodes, the stratigraphic positions of which are well constrained. In spite of episodic supply at the eruptive source, sediment accumulation is continuous at the resolution of the GRTS. This suggests that accumulation is controlled by basin subsidence rather than sediment supply. Sanidine dates support an age for the older boundary of marine magnetic anomaly 5 of about 10 Ma, as opposed to 11.12 Ma, suggested by the most recent results from Icelandic basaltic lava flows.

INTRODUCTION

The Baringo Basin of the Kenya Rift has long been a fruitful source of Neogene vertebrate fossils, including fossil hominoids [*Bishop and Chapman*, 1970; *Bishop et al.*, 1971; *Bishop and Pickford*, 1975; *Pickford*, 1975, 1978*a*, *b*]. Within the basin, a relatively complete sedimentary record spanning the last 20 Ma lies exposed in many disjunct outcrops. Temporal and stratigraphic correlations among the many exposures, although essential for the understanding of the vertebrate record, are only poorly constrained. The purpose of our research in the Baringo Basin is to establish a temporal framework based on K-Ar dating and paleomagnetic techniques. We present here a discussion of the paleomagnetic data from the type section of the Ngorora Formation as well as a discussion of the implications of our results for the calibration of the geomagnetic reversal time scale (GRTS).

GEOLOGICAL BACKGROUND

The Baringo Basin lies in the Kenya Rift between the equator and 1°N latitude (Figure 1). Regional mapping of the basin began as part of the East African Geological Research Unit directed by B. C. King, then of Bedford College (University of London). Early efforts focused on regional geochronology and structural aspects of the region and formed the basis for the stratigraphic sequence as it is understood today [*Martyn*, 1967; *McCall et al.*, 1967; *Bishop and Chapman*, 1970; *King and Chapman*, 1972; *Bishop and Pickford*, 1975].

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Paper number 4B5215.
0148-0227/85/004B-5215$05.00

The Miocene outcrops are dominated by thick sequences of lava flow units, including the Sidekh, Noroyan, Tiim, Ewalel, Kabarnet, and Kaparaina formations with interbedded sedimentary units including the Ngorora and Lukeino formations. The Ngorora Formation is defined as the strata overlying the Tiim phonolite and underlying the Ewalel phonolite. The following is a discussion of results from the type section of the Ngorora Formation at Kabarsero. Outcrops of Ngorora are shown in Figure 1.

LITHOSTRATIGRAPHY OF THE TYPE NGORORA FORMATION

The Kabarsero section of the Ngorora Formation (described by *Bishop and Pickford* [1975]) is composed of volcaniclastic and epiclastic sediments deposited on top of a weathered Tiim phonolite flow. It is well bedded on a millimeter to meter scale but displays substantial lateral facies variation, making lithostratigraphic correlations between isolated outcrops tenuous. Nearly all of the sediments are reworked pyroclastic material, deposited in lakes and streams or on floodplains. Basement-derived clasts are extremely rare, comprising less than 0.1 vol % of the exposed strata.

In the type area the formation is 420–460 m thick. At the base of the section the sediments are poorly sorted, volcaniclastic sandy siltstones and clays. Debris flows are common. The poorly sorted lower layers are overlain by cross-bedded tuffaceous fine sands and silts as well as an accretionary lapilli tuff. This unit is overlain by finely laminated clays and shales with minor channel-sand deposits. Superposed on these sediments is a sequence of clays interbedded with tuffaceous silts and sands. Near the top of the section is a thick unit of low-density white shales (diatuffs). These white shales consist primarily of altered volcanic ash but are in part diatomaceous with locally abundant fish debris. The white shale unit is punctuated by a single, poorly sorted, matrix-supported debris flow and is capped by a pumice lapilli, medium to coarse sandstone, known as the "buff tuff." The section between the

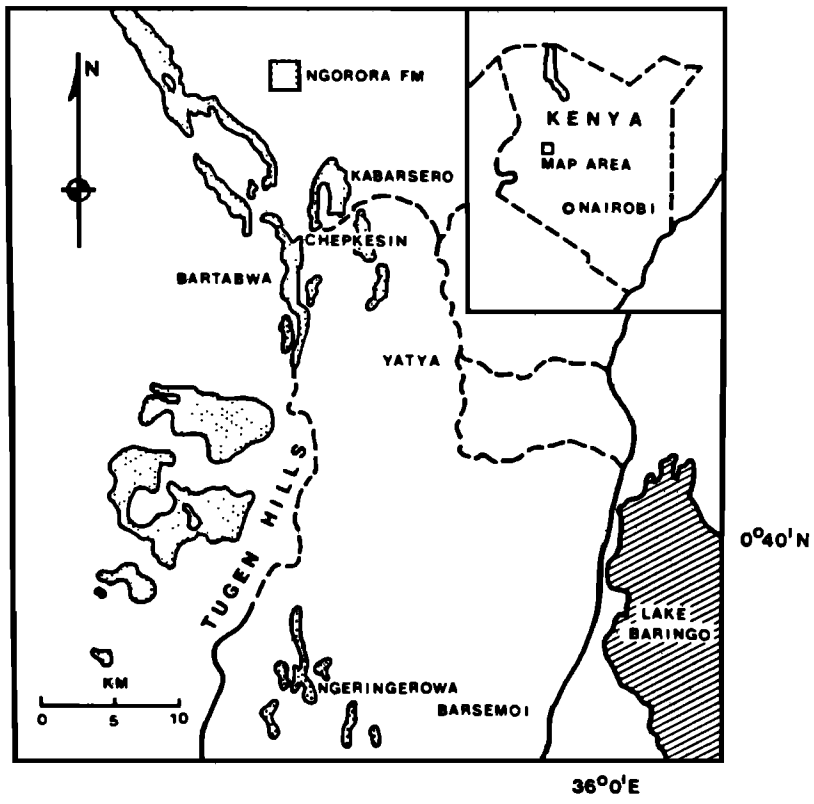


Fig. 1. Location of Baringo Basin. Stippled areas are outcrops of the Ngorora Formation. The type section is at Kabarsero.

"buff tuff" and the Ewalel phonolite is not exposed. The debris flow is used as a stratigraphic marker to correlate sections measured at two separate exposures within the type locality. The two sections are referred to as Kabarsero North and South (KN and KS).

PALEOMAGNETIC RESULTS

The Kabarsero section of the Ngorora Formation was first sampled in 1982 for paleomagnetic purposes, at an average density of one site (three samples) every 6 m for a total of 67 sites, using the method of *Johnson et al.* [1975]. In 1984, portions were resampled in order to refine the polarity stratigraphy. An additional 24 sites were taken. Our strategy for this preliminary study was to sample a range of lithologies in order to assess the suitability of the various rock types for paleomagnetic analysis. In general, sandy units were avoided in favor of finer-grained units. The sampled rock types include confetti clay (clay with millimeter-sized salmon pumice fragments), paper shales (finely laminated diatuff), claystones, siltstones, massive diatuff, and porcellanite.

The magnetic remanence of specimens collected in 1982 was measured using the Superconducting Technologies cryogenic magnetometer at Lamont-Doherty Geological Observatory. Stepwise thermal demagnetization was performed on 12 pilot specimens. The results for four typical specimens are shown in Figure 2. The variety of lithologies is reflected in the demagnetization behavior. Specimens NA1A and NA10A (Figures 2*a* and 2*b*, respectively) have a remanence carried by hematite, whereas NA13A and NA26A (Figures 2*c* and 2*d*, respectively) display blocking temperatures characteristic of magnetite. The polarity, however, was clearly defined by demagnetization to 300°C in all cases. Blanket demagnetization to 350°C was performed on all remaining specimens. Samples of six sites disintegrated during demagnetization were thus eliminated from analysis. The site means and precision parameters were calculated for the remaining sites. Random within-site directions, as determined by the test of *Watson* [1956], were caused by either truly random directions or a mixture of normal and reversely magnetized specimens within one site. The latter was particularly apparent in the vicinity of magnetozone boundaries. In either case, the resultant direction has little meaning; hence a further 22 sites proved unsuitable for paleomagnetic analysis and required additional sampling. Particularly disappointing were the results from the porcellanites and diatuffs, their being exceptionally weakly magnetized. Other lithologies proved well suited for paleomagnetic analysis.

The specimens collected in the 1984 field season were prepared and analyzed in the Scripps Paleomagnetic Laboratory (SPL) using the CTF Systems, Inc. three-axis cryogenic magnetometer. The measurement procedures in the SPL are as follows. The sample holder is measured first, followed by a baseline measurement. The specimen is measured in four orientations. The (very small) contribution of the sample holder is subtracted from the remanence measurements. A final baseline measurement is taken, and if drift has exceeded 30 pA m^2, the specimen is remeasured. Each measurement consists of an average of 100 separate measurements of the three axes. Standard

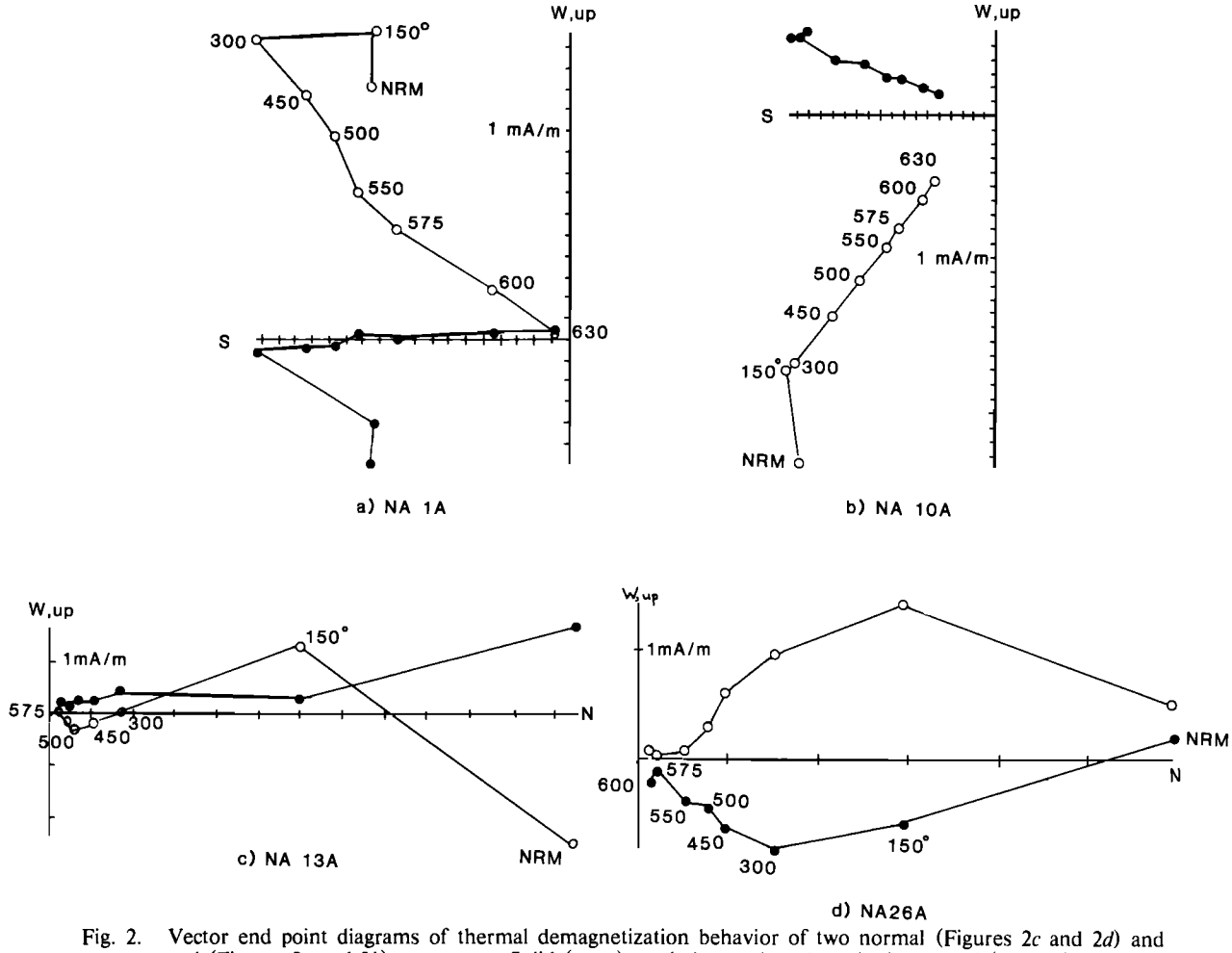


Fig. 2. Vector end point diagrams of thermal demagnetization behavior of two normal (Figures 2*c* and 2*d*) and two reversed (Figures 2*a* and 2*b*) specimens Solid (open) symbols are plotted on the horizontal (vertical) plane

deviations of the 100 measurements (an indication of instrument and/or sample noise) are not allowed to exceed 5 pA m^2. The circular standard deviation (CSD) of the four specimen vector measurements is calculated. The specimen is measured 3 times if the CSD exceeds 5° and 5 times for CSD's in excess of 10°. A vector mean of the multiple measurements is then calculated. By far the majority of measurements yield CSD's of less than 1°. These raw data are then corrected for sample and bedding orientation. The SPL cryogenic magnetometer is housed in a magnetically shielded room (ambient field of approximately 150 nT) in order to eliminate laboratory-acquired viscous remanence after demagnetization.

Because the samples collected in 1984 were taken from horizons that had previously given disappointing results, one specimen from each site was subjected to stepwise thermal demagnetization. The companions to specimens with smooth demagnetization behavior were then demagnetized to 500°C, and the site means were calculated. Of the 24 sites, 11 gave results that were statistically significant.

Paleomagnetic results are summarized in Table 1 and Figure 3. The mean normal and reversed directions are antipodal and are significantly different from the present field direction, suggesting some stability of remanence. These data are considered to reflect the polarity of the magnetic field at the time of sediment deposition. However, further investigations are needed to test this hypothesis. The nonrandom site mean directions (class A of *Tauxe and Opdyke* [1982]) were converted to virtual geomagnetic pole (VGP) positions using a site latitude of 1°N and longitude of 35.5°E. The VGP latitudes are plotted versus stratigraphic thickness in Figure 3, along with the K-Ar dates, lithostratigraphy, site locations, polarity interpretations, and preferred correlation to the GRTS of *LaBrecque et al.* [1977] ("LKC77") as recalculated by *Mankinen and Dalrymple* [1979] ("MD79b"). Our correlation to the GRTS was originally based on the constraints of isotopic dates published by *Bishop and Chapman* [1970] on the bounding lava flows of 9 and 12 Ma for top and bottom, respectively (no confidence limits supplied).

TABLE 1. Summary of Paleomagnetic Data

	Dec	Inc	*N*	*R*	$^a95$	*K*
Mean normals (geographic)	356	13	24	20.9	6.2	7
Mean reversed (geographic)	179	2	26	22.3	6	7

Site location: latitude, 1°N, longitude, 35.5°E; bedding orientation: strike, N5°E, dip., 14°W; present earth field: declination, 2°W, inclination, -20°; dipole field inclination: 2°.

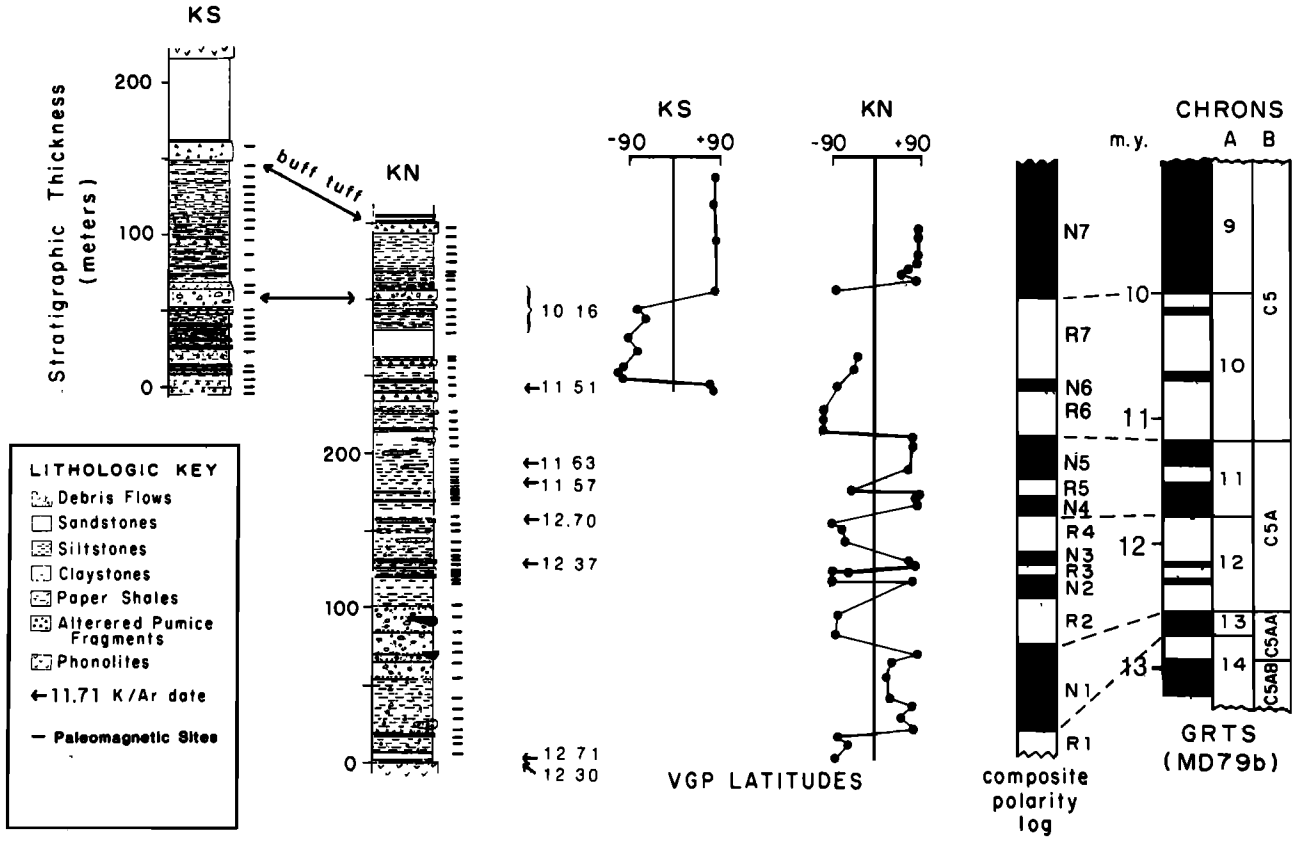


Fig. 3. Stratigraphic section with thickness in meters to left. K-Ar dates are those from Table 2. Columns labeled KN (Kabarsero North) and KS (Kabarsero South) are the stratigraphic sections as measured and described by us. Paleomagnetic sites are indicated by a tick mark to the right of lithology. Latitudes of the virtual geomagnetic pole (VGP) positions are plotted to the right of the site locations. These are interpreted in terms of polarity (normal/black, reversed/white). The magnetozones are labeled R1–N7 for reference in text. The magnetostratigraphy is correlated to the geomagnetic reversal time scale (GRTS) of *Mankinen and Dalrymple* [1979] (MD79b). Chron column B is the terminology of *Berggren et al.* [1985].

TABLE 2. K-Ar Dates From the Ngorora Formation at Kabarsero

KA	Sample	Material	Sample Weight	K ± 1σ (n), %	$^{40}Ar^*$ x 10^{-11} mol/g	$^{40}Ar^*$, %	Age ± 2σ, Ma	Location	Remarks
320R	RD-23	sanidine	2.2863	6.413 ± 0.08 (3)	11.74	84.5	10.52 ± 0.26	Kabarsero	
320R2	RD-23	sanidine	3.0342	6.413 ± 0.08 (3)	10.89	87.2	9.76 ± 0.24	10.16 ± 0.38 (1σ)	sample inhomogeneity gave variable K values and probably Ar as well
320R3	RD-23	sanidine	1.4781	6.413 ± 0.08 (3)	11.38	73.2	10.20 ± 0.26	10.16 ± 0.38 (1σ)	
441	RD-52	sanidine	2.5807	5.593 ± 0.05 (3)	11.20	90.3	11.51 ± 0.22	10.16 ± 0.38 (1σ)	
440	RD-55	sanidine	0.8381	5.742 ± 0.016 (3)	11.62	71.9	11.63 ± 0.11	10.16 ± 0.38 (1σ)	
438	RD-53	sanidine	3.2511	5.330 ± 0.022 (3)	10.72	84.9	11.57 ± 0.14	10.16 ± 0.38 (1σ)	
294	RD-51	sanidine	1.9877	5.480 ± 0.078 (3)	12.11	86.8	12.70 ± 0.38	10.16 ± 0.38 (1σ)	higher K variation
446	RD-54	sanidine	1.9770	5.004 ± 0.013 (3)	10.77	77.8	12.37 ± 0.10	10.16 ± 0.38 (1σ)	
292	RD-21	sanidine	3.3435	5.561 ± 0.01 (3)	12.30	90.3	12.71 ± 0.09	10.16 ± 0.38 (1σ)	
443	RD-50	whole rock phonolite	5.6673	5.144 ± 0.10 (3)	11.02	57.9	12.3 ± 0.50	10.16 ± 0.38 (1σ)	higher K variation alteration Ar loss?

The ^{40}K decay constant: $\lambda e + \lambda e' = 0.581 \times 10^{-11}$/yr., $\lambda\beta = 4.962 \times 10^{-11}$/yr.; $^{40}K/K = 1.167 \times 10^{-4}$.

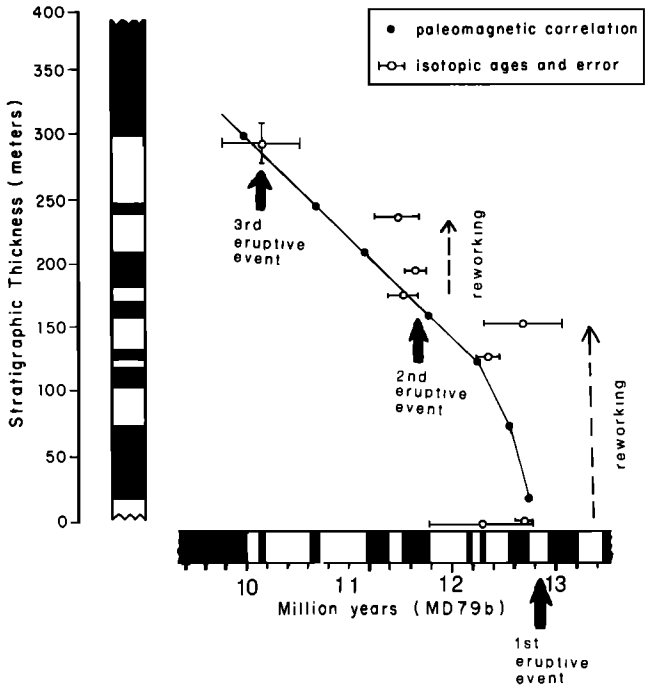


Fig. 4. The magnetostratigraphic pattern from Figure 3 is plotted versus the GRTS (MD79b). The solid line is the best fit through the points based on the paleomagnetic correlation shown in Figure 3. The open symbols represent K-Ar isotopic age dates of hand-picked sanidines from Table 2. See text for discussion.

These dates have since been supplemented by those presented in Table 2 and shown in Figure 3, which firmly establish the correlation.

The chron structure of the GRTS is well reproduced by the Ngorora magnetostratigraphy. One subchron, expected to fall within R7, is not represented.

DISCUSSION

Episodic Volcanism and Continuous Sediment Accumulation at Kabarsero

We have plotted our magnetostratigraphy against the MD79b time scale on Figure 4. We also have plotted our K-Ar isotopic ages (Table 2) for comparison. Our paleomagnetic correlation suggests an average sediment accumulation rate of 250 m/Ma for the sediments lying between the weathered Tiim phonolite and the 125-m level and a linear sediment accumulation rate of 88 m/Ma for the sediments lying above the 125-m level. These observations are in accord with the lithological character of the section (i.e., poorly sorted and coarse units in the bottom 125 m and moderately sorted and finer-grained units above the 125-m level). The continuous nature of sedimentation inferred from the magnetic ages contrasts with the clustering of the eight K-Ar ages. The K-Ar ages form three distinct age/stratigraphic interval groups: 12.6 ± 0.2 Ma (mean and standard deviation of three tuffs and the Tiim phonolite) for the interval 0–125 m; 11.6 ± 0.1 Ma (mean and standard deviation of three tuffs) for the interval 175–275 m; 10.16 ± 0.4 Ma (mean and standard deviation of three analyses on one tuff) for the interval above the 275-m level. Paleomagnetic and isotopic ages of reworked volcaniclastic material date two different sets of events: paleomagnetic ages give the time of sediment deposition and isotopic ages date the time of eruption. Our data indicate that the Kabarsero sediments were derived from volcanic materials having three distinct eruptive ages and that these materials were sampled sequentially without intermixing. This suggests that there was only one volcanically active source region, which provided volcaniclastics to a wide area in three major eruptive episodes. Subsequent to their deposition, the volcaniclastics were transported by surface processes to the paleo-Baringo Basin. It follows from this scenario that the isotopically constrained eruption ages must be equal to or older than the paleomagnetically constrained sedimentation ages. It is interesting to note that the sediments accumulated more or less continuously, within the resolution of the GRTS, despite both the episodic volcanism at the sediment source and the discontinuous deposition of deposits tens of meters thick. This strongly suggests that the initial high sediment accumulation rate reflects the rapid infilling of a topographic depression. The subsequent linear and lower sediment accumulation rate reflects basin subsidence at a moderate and constant rate.

Constraints on the Age of Marine Magnetic Anomaly 5

The data presented in the previous section constitute a rare opportunity to apply high-quality isotopic age data directly to the GRTS. We chose the *Mankinen and Dalrymple* [1979] recalculation of the *LaBrecque et al.* [1977] time scale for comparison with our data (Figure 3) and found excellent agreement. There are many successors to this time scale, however, and the validity, applicability, and accuracy of these various time scales merit some discussion. In the following section, we will briefly discuss the history of the calibration of the GRTS and then the implications of the Kenya data for the calibration of the GRTS.

Background to the GRTS. There are two fundamentally different records of the magnetic polarity history: the magnetostratigraphic record and the marine magnetic anomaly record. The magnetostratigraphic record consists of sediments or volcanics which were magnetized during (or soon after) deposition or eruption. These records are subject to abrupt changes in sediment accumulation rates or volcanicity and generally do not span long periods of time. The marine magnetic anomaly record results from the process of seafloor spreading which carries oceanic crust, magnetized at the time of formation, away from the mid-ocean ridges, forming lineated bands of reversely and normally magnetized crust. The seafloor spreading record is subject to distortions caused by changes in seafloor spreading rates and the episodic nature of volcanicity at mid-ocean ridges.

Geomagnetic reversal time scales are chronometric calibrations of polarity histories. They are derived from the two magnetic polarity records and rely on various assumptions as to the constancy of spreading rates, the reliability of isotopic age data, and the relationship of the age data to the magnetic polarity pattern. By far the best record of the polarity history is the marine magnetic anomaly record, complete from the present back into the Jurassic.

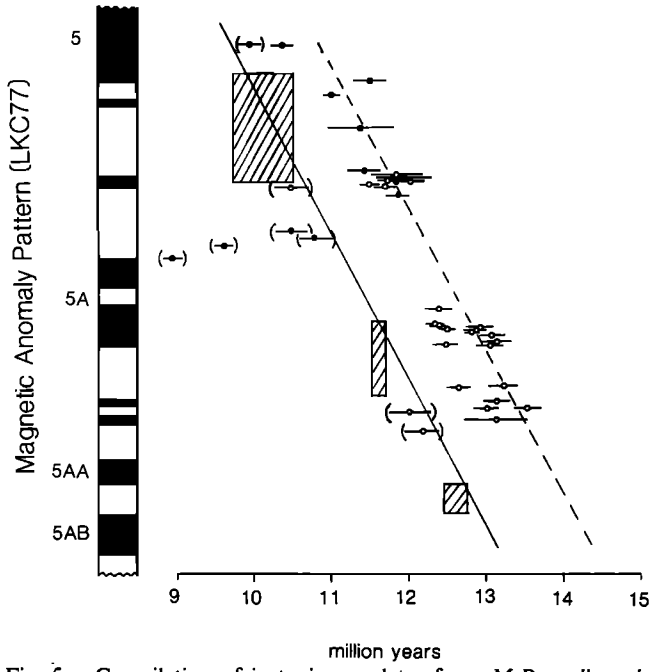


Fig. 5. Compilation of isotopic age dates from *McDougall et al.* [1984] and this paper with respect to the reversal pattern of *LaBrecque et al.* [1977]. Solid (open) circles are from the eastern (western) composite section of northwest Iceland. Data in parentheses were excluded in the original study as anomalous. Hatchured regions are the isotopic age data from Figure 4 as explained in text. The dashed line is the MKS84 time scale and the solid line is the LKC77* time scale.

Chronometric calibration of this record, however, is not entirely straightforward. Direct calibration would require dating of basalts recovered by deep-sea drilling from locations with clearly identified anomalies, but dates from oceanic basalts are affected by severe chemical alteration and are often unreliable. Calibration, then, relies on isotopic ages from stratigraphic sequences that have been correlated to a magnetic anomaly pattern.

There have been many attempts at calibrating the seafloor magnetic anomaly record. *Heirtzler et al.* [1968] ("HDHPL68") chose the Vema 20 South Atlantic magnetic anomaly record as the standard for the magnetic polarity history for the past 80 Ma. To derive their classic GRTS, they assumed a constant rate of seafloor spreading of 1.9 cm/yr (as calculated from the axial anomaly and anomaly 2A(o), dated at 3.37 Ma by *Doell et al.* [1966]).

Subsequent time scales have involved both refinements of the magnetic anomaly pattern and its chronologic calibration. *Talwani et al.* [1971] ("TWL71") revised the relative spacing of the first five numbered anomalies based on a detailed study of the Reykjanes Ridge in the North Atlantic; *Blakely* [1974] ("B74") did the same for the period between anomalies 5 and 6 based on a study in the North Pacific. *LaBrecque et al.* [1977] ("LKC77") synthesized several revisions of the anomaly pattern in order to derive the best estimate for the relative durations of the polarity units. They converted their ordinal sequence into years by interpolating between calibration points at anomalies 1(y), 2A(o) and 29(o) (0, 3.32, and 64.9 Ma, respectively). These dates are based on the old decay rates of ^{40}K to ^{40}Ar, recently revised by *Steiger and Jaeger* [1978].

Mankinen and Dalrymple [1979] presented two time scales based on the new potassium argon decay systematics and potassium isotope abundances. The first time scale (MD79a) spans the period 0–5 Ma and is based on revised age estimates for the Chron boundaries Brunhes/Matuyama, Matuyama/Gauss, and Gauss/Gilbert. The boundary ages are derived from hundreds of age determinations on lavas whose magnetic polarities are known. The second time scale (MD79b) is a recalculation of LKC77 whereby ages for each reversal are estimated by using a correction factor based on the new decay constants of ^{40}K as reported by *Steiger and Jaeger* [1978]. Since this correction factor is nonlinear, nonlinear spreading rates result as an artifact of the procedure. Perhaps a more "correct" approach would be to calculate a regression line through the present and the calibration points provided by MD79a and then apply a linear correction factor to LKC77. The calculated linear correction factor is

$$\text{new date} = (\text{old date} \times 1.02376) + 0.0068$$

We will hereafter refer to this recalibration as LKC77*.

Ongoing work in Iceland recently resulted in a revised GRTS for the time period Chron C4A (7) to C5AA (13) [*McDougall et al.*, 1984] ("MKS84"). By combining K-Ar isotopic ages with polarity determinations of basaltic lavas, the MKS84 time scale becomes an attempt to extend the direct calibration of the GRTS beyond 5 Ma.

Comparison with the Ngorora data set. The data set of *McDougall et al.* [1984] overlaps in time with the data presented here and the two can be directly compared. In Figure 5, we have plotted the isotopic ages from Iceland (eastern and western composites) against the uncalibrated magnetic polarity pattern of *LaBrecque et al.* [1977] according to the correlation preferred by *McDougall et al.* [1984]. Also plotted are the mean sanidine ages from Kenya for each eruptive episode, as correlated to the magnetic anomaly pattern in Figure 4. Stratigraphic limits for each episode are determined as follows. The lowermost episode (12.5 Ma) is assumed to be time equivalent with the Tiim phonolite which is known to correlate with Chron 5AB-r (Figure 3). The middle episode (11.6 Ma) is constrained to lie between the uppermost tuff belonging to the Tiim episode (first) and the lowermost tuff belonging to the second episode. The upper episode (10.2 Ma) is similarly constrained.

The data shown in Figure 5 indicate that (1) The K-Ar dates for euhedral sanidines from Kenya (Table 2) are within the scatter of the Icelandic K-Ar dates, (2) the two time scale models, MKS84 and LKC77*, bracket 90% of the K-Ar dates, and (3) LKC77* passes through not only the Kenya eruption ages but also through an additional four points from Iceland, which were excluded by *McDougall et al.* [1984] as "anomalous."

The chronometric calibration of MKS84 is based on the premise that it is more likely that the Icelandic basalts have suffered argon loss (resulting in ages that are too young) than potassium loss or incomplete outgassing of inherited Argon (resulting in ages that are too old).

Our preference of LKC77* is based on (1) our interpretation of the age groups as resulting from reworking of sanidines from discrete eruptive episodes and (2) an

assumption that argon loss from euhedral sanidines has been minimal.

LKC77* (supported by the Kenya data) and MKS84 (derived from Icelandic data) differ by more than 10%, far greater than the analytical error of the age determinations (typically 2%). *McDougall et al.* [1984] point out that estimates of the age of the older boundary of anomaly 5 based on various subsets of Icelandic data, range from 10.3 to 11.12 Ma, a difference of more than 7%. They attribute the discrepancy to systematic loss of radiogenic argon from slightly altered, low-potassium basalts. Other possible causes for the discrepancy between the Kenya and Icelandic data sets include (1) a miscorrelation of the magnetostratigraphy to the magnetic anomaly pattern in one or both studies, (2) loss of radiogenic argon from the Kenya sanidines, (3) loss of potassium during alteration of Icelandic basalts, (4) incomplete outgassing of argon during eruption of the Icelandic lavas, and (5) incorporation of argon in the Icelandic basalts by hydrothermal alteration.

It is possible that parts of the Icelandic magnetostratigraphic sections are not correctly correlated to the magnetic anomaly pattern, but the identification of marine magnetic anomaly 5 seems incontrovertible. The correlation of the Kenya magnetostratigraphy (Figure 3) is excellent. The discussion, therefore, centers on the reliability of the isotopic ages.

K-Ar isotopic systematics may be disturbed by a number of alteration processes affecting the parent/daughter ratios and/or the content of radiogenic argon. Loss of radiogenic argon is the most cited process which decreases the K-Ar age of a geologic sample. However, diffusion rates of argon out of sanidine are too slow to allow escape of argon from the Kenya sanidines; on the basis of the thermal demagnetization behavior of the Ngorora specimens, the Kenya sediments have not experienced temperatures in excess of 150°C (Figure 2*b*). For this reason, we believe that the sanidine ages reported in Table 2 are reliable estimates of the eruption ages.

There are two processes which may increase the apparent age of basalts: (1) basalt may have excess argon from incomplete outgassing caused by quenching at eruption, and (2) basalt can acquire radiogenic argon from hydrothermal solutions [e.g., *Lanphere and Dalrymple*, 1976]. If a rock contains "excess" radiogenic argon relative to its potassium content, the age appears too old. It has been repeatedly shown that K-Ar systematics of basalts can easily be affected by alteration and that extreme care in selecting the least altered samples results in a reduced scatter of the K-Ar age dates [e.g., *McDougall and Schmincke*, 1976]. Basalts from Iceland have been reported to show signs of alteration [*McDougall et al.*, 1984]. We therefore believe that our sanidine dates are more reliable than the whole rock basalt dates from Iceland. We interpret the Kenya data, therefore, to support a younger age estimate of approximately 10 Ma for the older boundary of marine magnetic anomaly 5 as predicted by LKC77*.

Paleomagnetism and the art of correlation. Perhaps the most significant lesson to be learned from the previous discussion is that paleomagnetic data, when combined with isotopic age data, allow an unprecedented precision in stratigraphic correlation. Isotopic ages may not always give true ages as revealed by paleomagnetic correlation. K-Ar dates allow a resolution of perhaps 10%. Paleomagnetic techniques cannot stand without age estimates provided by isotopic or biostratigraphic data. The full potential of both isotopic and magnetostratigraphic techniques is therefore realized when the two are used together.

CONCLUSIONS

The principal results of our paleomagnetic, K-Ar, and lithologic studies may be summarized as follows:

1. The Ngorora sediments are well suited for paleomagnetic analysis. Stable directions are isolated by demagnetization to 300° or 350°C. The carrier of remanence is either magnetite or hematite, depending on lithology. The consistency of the data set strongly suggests that the remanence was acquired penecontemporaneously to deposition. However, detailed rock magnetic studies are required to substantiate further this conclusion.

2. Sanidine K-Ar dates fall into three groups of constant age, corresponding to three discrete stratigraphic intervals. The groups, 12.5, 11.6, and 10.2 Ma, are separated by lacunae of approximately 1 m.y. duration. This distribution is interpreted to be the result of reworking of sanidines from three discrete episodes of eruption. The K-Ar dates, therefore, do not apply to the age of sediment deposition except at the base of each stratigraphic interval.

3. Despite episodic activity in the volcanic source region, accumulation of volcaniclastic sediment is continuous, within the resolution of the magnetostratigraphic technique. This suggests that sediment accumulation was controlled primarily by basin subsidence as opposed to sediment availability.

4. Tying our K-Ar dates to the geomagnetic reversal sequence allows comparison with other dated sequences. The results presented in this paper and those presented by *McDougall et al.* [1984] yield chronometric calibrations of the geomagnetic reversal time scale differing by approximately 1 m.y. We feel that our calibration, based on sanidine K-Ar dates rather than on whole rock basalt K-Ar dates, is more reliable. Our results indicate an age of approximately 10 Ma for the older boundary of marine magnetic anomaly 5 rather than 11.12 Ma as suggested by *McDougall et al.* [1984].

5. Magnetostratigraphy, when used in conjunction with isotopic techniques, provides a means of stratigraphic correlation with an unprecedented resolution. The combined results of paleomagnetism and isotopic dating are far better than either technique used alone.

Acknowledgments. We are pleased to acknowledge the members of the Baringo Paleontological Research Project: D. R. Pilbeam, J. Barry, A. K. Behrensmeyer, B. Brown, A. Hill, L. Jacobs, B. F. Jacobs, N. M. Johnson and Steve Ward. Special thanks go to D. V. Kent, J. Flynn, and J. LaBrecque for many stimulating conversations and to D. R. Pilbeam, E. H. Lindsay, J. E. T. Channell, and G. Mead for critical readings of the manuscript. Warmest thanks go to Dave R. Clark for invaluable laboratory assistance and Sally McBreardy for drafting. Many thanks to Mary Beth Hiller for typesetting the manuscript. This work was supported by National Science Foundation (NSF) grants EAR82-12763 and EAR83-05751 and grants from the Leakey Foundation and the Foundation for Research into the Origin of Man to the senior author. Field support was made available by NSF grant BNS81-40818 and a grant from the Louise Brown Foundation to D. Pilbeam and A. Hill. Funds for equipment for

the Scripps Paleomagnetic Laboratory were kindly provided by the W. M. Keck Foundation.

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(Received August 1, 1984;
revised December 26, 1984;
accepted January 24, 1985.)